

What was the most challenging part?

The hardest part of this assignment wasn't writing the model or the Dockerfiles - it was getting the actual Kubernetes training Job to run correctly, because two separate bugs only showed up once real data hit the pipeline, not during simple builds or health checks.

The first issue was a NumPy/PyTorch version conflict. Both requirements files pinned torch==2.2.2 but never pinned numpy, so pip quietly installed NumPy 2.x alongside it. The Docker image built fine, and the serving container's /health check even passed, because none of that touches real data. But the moment the training Job tried to load and transform an actual CIFAR-10 image, it crashed with "RuntimeError: Numpy is not available" - torch 2.2.2's compiled extensions expect NumPy 1.x's ABI. The fix was one line (numpy==1.26.4 in both requirements files), but finding it took an actual end-to-end run, since Docker-level testing alone never exercised that code path.

The second issue was sneakier: the training Job would sit there "Running" for 40+ minutes without producing a single log line, even though the pod's CPU usage looked fully active. It wasn't crashing or restarting, just seemingly stuck. Digging into /proc inside the pod showed the process was genuinely burning CPU the whole time, not idle - so it wasn't a real deadlock. The actual cause was that PyTorch defaults its internal thread pool to the machine's total CPU count (12, in this case), but the Job's resource limit only allowed it 2 CPU-cores worth of scheduling time. Twelve threads fighting over two cores' worth of quota meant most of the CPU time went to context-switching overhead instead of actual computation. Setting OMP_NUM_THREADS and MKL_NUM_THREADS to 2 (matching the CPU limit) fixed it - the same benchmark that hadn't finished after 5+ minutes at 12 threads finished predictably once capped correctly.

A smaller, more mundane challenge was just how slow CPU-only training is for something like ResNet-18 - a single epoch on the full CIFAR-10 dataset took roughly an hour even after fixing the threading issue, which made iterating on the Kubernetes manifests slow and required patience (and a scaled-down config) to validate the mechanics quickly before committing to a full run.

Overall, the biggest lesson was that Docker build success and a passing health check don't prove a pipeline actually works - only running it against real data on real infrastructure surfaces the failures that matter.

```
Terminal Shell Edit View Window Help
arpit.singh --zsh.kubectl -- 244x63
...age8-api-all-logsback/batchman --zsh ... --zsh.kubectl ... --zsh ... --zsh ... --zsh ... --zsh ... --zsh ... --zsh ...
curl http://localhost:8080/health
curl -X POST http://localhost:8080/predict -F "image=@tmp/test_image.png"

Switched to context "kind-mlops-pytorch-pipeline".
NAME      STATUS      COMPLETIONS  DURATION  AGE
model-training  Complete  1/1          5h0m      20h
Files already downloaded and verified
Files already downloaded and verified
{"epoch": 1, "train_loss": 1.3657, "train_accuracy": 0.499, "val_loss": 1.159, "val_accuracy": 0.6832}
{"event": "checkpoint_saved", "path": "/app/checkpoints/classifier_v1.pt"}
{"event": "training_complete", "best_val_loss": 1.159}
NAME      READY  STATUS      RESTARTS  AGE
model-serving-64d79654d7-25rmv  1/1    Running    0          20h
model-serving-64d79654d7-gkx6b  1/1    Running    0          20h
model-training-9m9g6           0/1    Completed  0          20h
Name:      model-serving
Namespace:  ml-training
CreationTimestamp:  Sat, 29 Aug 2026 16:25:03 +0530
Labels:     <none>
Annotations:  deployment.kubernetes.io/revision: 1
Selector:    app=model-serving
Replicas:    2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType: RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 0 max unavailable, 1 max surge
Pod Template:
Labels:  app=model-serving
Containers:
  serving:
    Image:      mlops-serve:v1
    Port:       8080/TCP
    Host Port:  0/TCP
    Limits:
      cpu:      1
      memory:  2Gi
    Requests:
      cpu:      500m
      memory:  1Gi
    Liveness:   http-get http://:8080/health delay=0s timeout=1s period=10s #success=1 #failure=3
    Readiness:  http-get http://:8080/health delay=15s timeout=1s period=5s #success=1 #failure=3
    Environment:  <none>
    Mounts:
      /app/checkpoints from checkpoints-volume (ro)
      /app/configs from config-volume (ro)
Volumes:
  checkpoints-volume:
    Type:  PersistentVolumeClaim (a reference to a PersistentVolumeClaim in the same namespace)
    ClaimName:  checkpoints-pvc
    ReadOnly:  true
  config-volume:
    Type:  ConfigMap (a volume populated by a ConfigMap)
    Name:  training-config
    Optional:  false
  Node-Selectors:  <none>
  Tolerations:    <none>
Conditions:
  Type      Status  Reason
  --      -
  Available  True    MinimumReplicasAvailable
  Progressing  True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   model-serving-64d79654d7 (2/2 replicas created)
```

```
Terminal Shell Edit View Window Help
arpit.singh -- zsh • kubecti -- 244x63
...age8-api-all-/-logback/batchman -- zsh
-- zsh • kubecti
-- zsh
...
-- arpit-singh@stage8-ic-api -- zsh
...e7-api-all-/-logback/balance-api -- zsh
+

{"event": "training_complete", "best_val_loss": 1.158}
NAME          READY   STATUS    RESTARTS   AGE
model-serving-64d79664d7-26rsv  1/1     Running   0           20h
model-serving-64d79664d7-2kcrb  1/1     Running   0           20h
model-training-9mgg6           0/1     Completed 0           20h

Name:          model-serving
Namespace:     ml-training
CreationTimestamp: Sat, 29 Aug 2026 16:25:53 +0530
Labels:        <none>
Annotations:   deployment.kubernetes.io/revision: 1
Selector:      app=model-serving
Replicas:      2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType:  RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 0 max unavailable, 1 max surge
Pod Template:
  Labels:  app=model-serving
  Containers:
    serving:
      Image:      mlops-serve:v1
      Port:       8080/TCP
      Host Port:  0/TCP
      Limits:
        cpu:      1
        memory:  2Gi
      Requests:
        cpu:       500m
        memory:    1Gi
      Liveness:     http-get http://:8080/health delay=8s timeout=1s period=10s #success=1 #failure=3
      Readiness:    http-get http://:8080/health delay=15s timeout=1s period=5s #success=1 #failure=3
      Environment: <none>
      Mounts:
        /app/checkpoints from checkpoints-volume (ro)
        /app/configs from config-volume (ro)
  Volumes:
    checkpoints-volume:
      Type:      PersistentVolumeClaim (a reference to a PersistentVolumeClaim in the same namespace)
      ClaimName: checkpoints-pvc
      ReadOnly:  true
    config-volume:
      Type:      ConfigMap (a volume populated by a ConfigMap)
      Name:      training-config
      Optional:   false
    Node-Selectors:  <none>
    Tolerations:     <none>
Conditions:
  Type              Status  Reason
  ----              -
  Available          True    MinimumReplicasAvailable
  Progressing        True    NewReplicaSetAvailable
  OldReplicaSets:    <none>
  NewReplicaSet:     model-serving-64d79664d7 (2/2 replicas created)
  Events:             <none>
[2] 16005
Unable to listen on port 8080: Listeners failed to create with the following errors: [unable to create listener: Error listen tcp4 127.0.0.1:8080: bind: address already in use unable to create listener: Error listen tcp6 [::]:8080: bind: address already in use]
error: unable to listen on any of the requested ports: [{8080 8080}]
[2] + exit 1 kubecti port-forward svc/model-serving 8080:80 -n ml-training
Handling connection for 8080
{"status": "ok"}Handling connection for 8080
{"predictions":[{"class": "ship", "probability": 0.19526}, {"class": "truck", "probability": 0.16377}, {"class": "automobile", "probability": 0.113274}, {"class": "cat", "probability": 0.183262}, {"class": "horse", "probability": 0.192552}, {"class": "airplane", "probability": 0.085738}, {"class": "bird", "probability": 0.071926}, {"class": "deer", "probability": 0.06473}, {"class": "frog", "probability": 0.051797}, {"class": "dog", "probability": 0.047685}]}
arpit.singh@IC-18044 ~ %
```